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18.085 Applied Mathematics

Text: Introduction to Applied Math by G. Strang

Instructor: Lizhen Ji at room 2-269 and extension x3-3778

In this course, there will be one middle term exam and one final exam. The date of the midterm exam is around early April and the exact date will be announced in class a couple of weeks before it. Homeworks account for 15% of the final grade, the midterm exam 30% and the final 55%.

We will cover the following topics. The numbers listed after each topic are the corresponding homeworks. The homeworks are due on Mondays following the lectures.

1. Review of linear algebra and incidence matrices for graphs (1.6.9, 1.6.32, 1.6.2, 1.6.3)
2. Gaussian elimination and LU factorization (1.2.1, 1.2.4, 1.2.6)
3. Positive definite matrices (1.3.1, 1.3.3, 1.3.7)
4. Minimum principle and least square (1.4.1, 1.4.3, 1.4.5)
5. Eigenvalues and dynamical systems (1.5.1, 1.5.3, 1.5.5, 1.5.6) 1.4.11
6. Equilibrium equations (2.1.2, 2.1.3, 2.1.6, 2.1.8)
7. Constraints and Lagrange multiplier (2.2.1, 2.2.3, 2.2.5)
8. Electrical networks (2.3.2, 2.3.3)
9. Structures in equilibrium (2.4.1, 2.4.5, 2.4.8, 2.4.15)
10. Equilibrium in the continuous case and perturbation theory (3.1.1, 3.1.3, 3.1.5, 3.1.13)
11. Differential equations and calculus of variations (3.2.1, 3.2.2, 3.2.6, 3.6.1, 3.6.2, 3.6.15)
12. Laplace equation and potential flow in dimension 2 (3.3.1, 3.3.2, 3.3.14, 3.3.17, 3.3.18)
13. Vector calculus in dimension 3 (3.4.3, 3.4.4, 3.4.5, 3.4.12)

14. Fourier series and orthogonal expansions (4.1.1, 4.1.2, 4.1.4, 4.1.5, 4.1.16, 4.1.17, 4.1.26)

15. Discrete Fourier series and convolution (4.2.1, 4.2.2, 4.2.3, 4.2.5, 4.2.11)

16. Fourier integrals (4.3.1, 4.3.2, 4.3.3, 4.3.4, 4.3.10)

17. Complex variables and the Laplace equation (4.4.1, 4.4.2, 4.4.3, 4.4.5, 4.4.7)

18. The fast Fourier transformation

19. Initial value problems for ODE and their stabilities

20. Heat equations

February 11, 1994
18.445. First assignement.
Work out all the problems and write down your answers in neat form, in ink or preferably typewritten form. Each problem is worth 10 points, for a total of 100 points. Probleams are marked E = easy, M = medium H = hard.

Consider Maxwell-Boltzmann statistics, where the boxes are numbered 1,2,...,n.
Condition the sample space so that all occupation number equal 0 or 1.
Let $X_{(1)}$, $X_{(2)}$, ..., $X_{(k)}$ be the number of the first, second,...,k-th occupied box . Let L_1, L_2, ...,L_{k+1}, equal, respectively, $X_{(1)}$, $X_{(2)} - X_{(1)}$, $X_{(3)} - X_{(2)}$, ..., $n - X_{(k)}$ (these are the distances between successively occupied boxes).

1.M Compute the joint distribution of any subset of $X_{(1)}$, $X_{(2)}$, ..., $X_{(k)}$.
2. M Compute the joint distribution of any subset of L_1, L_2, ...,L_{k+1}.
3. M What is the probability that no two adjacent boxes are occupied?
4. E Compute the expectation of $X_{(1)}$, $X_{(2)}$, ..., $X_{(k)}$ and of L_1, L_2, ...,L_{k+1}.
5.E Define analogous random variables for Fermi-Dirac statistics, and compute the analogous joint probabilities aas in problems 1,2,3 and 4.
6.H Let Y_n (n = 1,2,3,...) be i.i.d. random variables taking the values ±1 with probability 1/2. What is the probability that the series

$$\Sigma\, Y_n/n$$

converges?
7.E State precisely the property of countable additivity of a probability function. Show by a typical example that this property is essential in computing probabilities of events.
8.E State precisely the Borel-Cantelli lemma, and give a full proof, without omitting any detail (warning: the proof given in the notes given out in class omits several crucial details.)
9.E Consider the following argument:
The event that in tossing a fair coin the pattern 011 "ever" occurs before the pattern 100 is a tail event. Therefore, by the zero-one law, its probability is either 0 or 1.
This argument is flawed. Restate it in correct form.
10. E Give the statement and full proof of the law of large numbers. Do not omit any details (points will be taken off for details left out.)